

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)	Number of [free] <u>electrons</u> per unit volume Accept electron density	1			1		
		(ii)	Volume = vtA or lA (1) can be obtained from clear diagram No. of [free] electrons = $vtAn$ or lAn (1) Total charge = $vtAne$ or $lAne$ (1) Current = $\frac{vtAne}{t}$ or $\frac{lAne}{t}$ (1) Accept alternatives e.g. in 1 second or per second	4			4	1	
		(iii)	Substitution and rearrangement: i.e. $v = \frac{I}{nAe} = \frac{1.8}{9.0 \times 10^{28} \times 2.8 \times 10^{-8} \times 1.6 \times 10^{-19}}$ (1) allow errors in powers and use of diameter for radius for first mark Answer = $4.4 \times 10^{-3} \text{ [m s}^{-1}\text{]}$ (1)		2		2	2	
	(b)	(i)	Same pd across 12Ω and 24Ω or 2.4 V across both 12Ω and 24Ω (1) I through 24Ω is $\frac{1}{2}$ that through 12Ω or 0.1 A through 24Ω (1) Parallel branch currents add to 0.3 A or $0.2 \text{ A} + 0.1 \text{ A} = 0.3 \text{ A}$ (1)		3		3		
		(ii)	$\frac{1}{R_p} = \frac{1}{12} + \frac{1}{24}$ to give $R_p = 8 \text{ } [\Omega]$ (1) $40 + 8$ ecf = $48 \text{ } [\Omega]$ (1) $V = IR = 0.3 \times 48$ ecf = 14.4 [V] (1) Alternatively: pd across $40 \Omega = IR = 0.3 \times 40 = 12 \text{ [V]}$ (1) pd across parallel branch = 2.4 [V] (1) Total pd = $12 + 2.4 = 14.4 \text{ [V]}$ ecf (1)		3		3	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	For $40\ \Omega$, power $[= I^2R = 0.3^2 \times 40] = 3.6\ \text{[W]} (1)$ Power from parallel arrangement $= [I^2R = 0.3^2 \times 8] = 0.72\ \text{[W]} (1)$ Alternative for the first 2 marks Correct use of IV or $\frac{V^2}{R}$ Third mark: $\frac{3.6}{0.72} [= 5]$ or $\frac{0.72}{3.6} = [\frac{1}{5}] (1)$ Alternative: $\frac{2.4}{12}$ or $\frac{8}{40} (1)$ $= \frac{1}{5} (1)$ Because current is the same or equivalent (1)		3		3	3	
			Question 2 total	5	11	0	16	8	0